Square derivatives: singular modulo 2 or over Q_2?

Unipotent center
$A = [[1, 1], [0, 1]]$
D is invertible over Q_2

Diagonal center
$A = \mathrm{diag}(1, -1)$
D is singular over Q_2

Scalar degree valuation: 1
Inverse denominator exponent: 2
Use $m > s = 2$

2	0
0	-2

$D(E_{ij})$ = shown scalar × E_{ij}

An invertible matrix center can have a singular power derivative.